



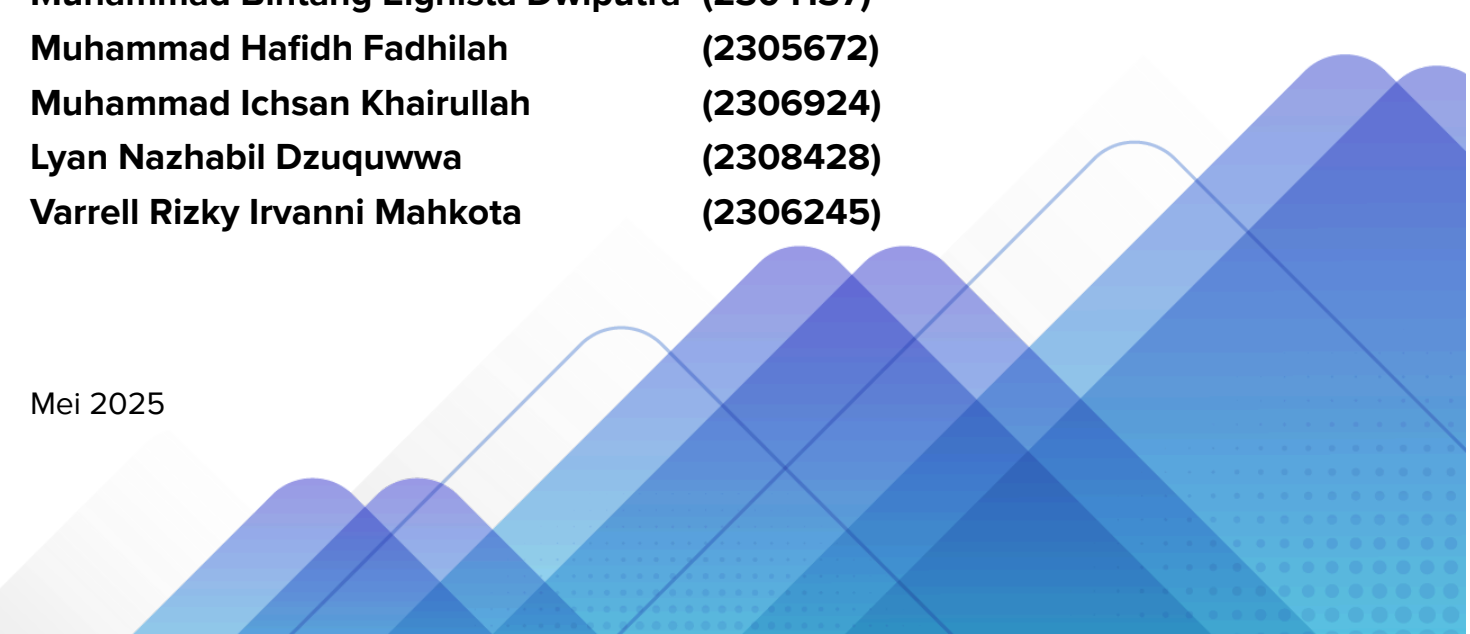
SPEKIFIKASI KEBUTUHAN DATA DASHBOARD

Big Data

Anggota:

Muhammad Bintang Eighista Dwiputra	(2304137)
Muhammad Hafidh Fadhilah	(2305672)
Muhammad Ichsan Khairullah	(2306924)
Lyan Nazhabil Dzuquwwa	(2308428)
Varrell Rizky Irvanni Mahkota	(2306245)

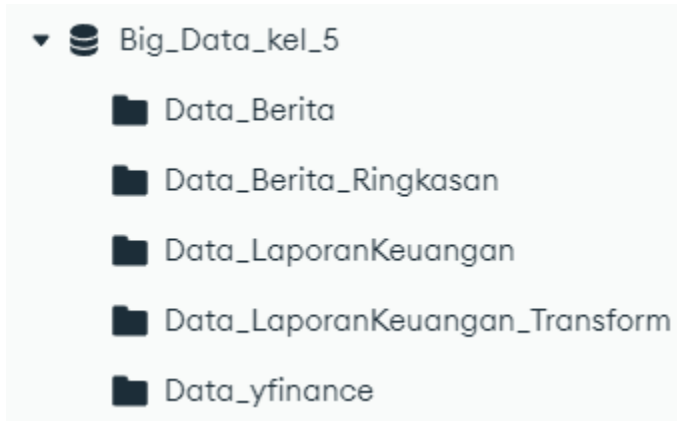
Mei 2025



Link Repository : <https://github.com/llyessin12/Tugas3Kelompok5>

1. Sumber Data

Sumber data untuk dashboard disimpan di database MongoDB atlas



Terdapat 5 koleksi yang digunakan:

- Data_Berita: menyimpan data berita hasil scraping dari web iqplus

```
_id: ObjectId('68204f9b3ac56e9bf654f91f')
Emiten : "BBRI"
Date : "09/05/25 - 14:18"
Title : "BBRI: KUALITAS KREDIT BRI SEMAKIN MEMBAIK DENGAN PENCADANGAN KUAT"
Link : "http://www.iqplus.info/news/stock_news/bbri-kualitas-kredit-bri-semaki..."
```

- Data_LaporanKeuangan: menyimpan data laporan keuangan hasil scraping dari web idx

```
_id: ObjectId('67c93725860f03985e68a36f')
company : "ADRO"
timestamp : "2025-03-06T18:11:16.551593"
  taxonomy : Object
  xbrl_data : Object
```

- Data_yfinance: menyimpan data saham hasil scraping dari Yahoo Finance

```
_id: ObjectId('67c6a2174ee732cea86e88db')
Date : "07/03/2024 - 00:00"
Open : 6325.8319670025
High : 6446.0949321546
Low : 6325.8319670025
Close : 6397.9897460938
Volume : 917900
Dividends : 0
Stock Splits : 0
emiten : "AALI.JK"
```

- Data_Berita_Ringkasan: menyimpan data berita yang sudah dirangkum menggunakan LLM

```
_id: ObjectId('680155d1415b696fddb01516')
Emiten : "AALI"
Date : "14/01/25 - 13:50"
Judul : "AALI: JUNJUNG TINGGI INTEGRITAS, ASTRA AGRO RAIH ANUGERAH TEMPO DAN TI..."
Link : "http://www.iqplus.info/news/stock_news/aali-junjung-tinggi-integritas-..."
Ringkasan : "PT Astra Agro Lestari Tbk (AALI) menerima penghargaan Indeks Integrita..."
uploaded_at : 2025-04-18T02:26:09.117+00:00
```

- Data_LaporanKeuangan_Transform: menyimpan data laporan keuangan yang sudah ditransform dari format .xbrl ke dalam format .json

```
_id: ObjectId('68219b3733594724975365c0')
EntityName : "Alamtri Resources Indonesia Tbk"
EntityCode : "ADRO"
SalesAndRevenue : 1084004000
GrossProfit : 867681000
ProfitFromOperation : 0
ProfitLoss : 1854878000
CashAndCashEquivalents : 3311232000
Assets : 10472711000
ShortTermBankLoans : 0
LongTermBankLoans : 404361000
EquityAttributableToEquityOwnersOfParentEntity : 6772664000
NetCashFlowOp : 1152758000
NetCashFlowInv : -582426000
NetCashFlowFin : -1333690000
PeriodOfFinancialStatementsSubmissions : "Tahunan / Annual"
DateOfBoardOfDirectorsStatementLetter : "2025-02-28"
CurrentPeriodStartDate : "2024-01-01"
CurrentPeriodEndDate : "2024-12-31"
PriorYearEndDate : "2023-12-31"
PriorPeriodStartDate : "2023-01-01"
PriorPeriodEndDate : "2023-12-31"
Prior2YearEndDate : "2022-12-31"
```

2. API Endpoints

API system diimplementasikan pada port 5000 dengan tiga endpoint utama:

2.1 Laporan Keuangan

GET /api/financial/<emiten>

Response:

```
[
  "EntityName": "string",
  "EntityCode": "string",
  "SalesAndRevenue": "number",
  "GrossProfit": "number",
```

```
"ProfitFromOperation": "number",
"ProfitLoss": "number",
"CashAndCashEquivalents": "number",
"Assets": "number",
"ShortTermBankLoans": "number",
"LongTermBankLoans": "number",
"EquityAttributableToEquityOwnersOfParentEntity": "number",
"NetCashFlowOp": "number",
"NetCashFlowInv": "number",
"NetCashFlowFin": "number",
"PeriodOfFinancialStatementsSubmissions": "string",
"DateOfBoardOfDirectorsStatementLetter": "string",
"CurrentPeriodStartDate": "string",
"CurrentPeriodEndDate": "string",
"PriorYearEndDate": "string",
"PriorPeriodStartDate": "string",
"PriorPeriodEndDate": "string",
"Prior2YearEndDate": "string"
]
```

2.2 Data Saham

GET /api/stock/<emiten>?period=<periode, seperti monthly, daily, weekly, quarterly, 1y, 3y, 5y>

Response:

```
[
  "symbol": "string",
  "open": "number",
  "high": "number",
  "low": "number",
  "close": "number",
  "volume": "number",
  "dividends": "number",
  "stock_splits": "number",
  "date": "string"
]
```

2.3 Berita

GET /api/news/<emiten>?limit=<limit>&skip=<skip>

Response:

```
[  
  emiten: string  
  date: string  
  title: string  
  link: url  
  content: string  
]
```

2.4 Ringkasan Berita

GET /api/news/summary/<emiten>?limit=<limit>&skip=<skip>

Response:

```
[  
  emiten: string  
  date: string  
  title: string  
  link: url  
  Ringkasan:string  
]
```

3. Konfigurasi Database

Local MongoDB Setup:

Database Name: Big_Data_kel_5

Port: 27017

Authentication: Enabled

Collections:

- Data_LaporanKeuangan
- Data_LaporanKeuangan_Transform
- Data_yfinance
- Data_Berita
- Data_Berita_Ringkasan

Environment Variables (.env):

```
# Database Configuration
MONGODB_DATABASE_NAME=Big_Data_kel_5

# Collections
COLLECTION_FINANCIAL_REPORTS=Data_LaporanKeuangan_Transform
COLLECTION_YFINANCE_DATA=Data_yfinance
COLLECTION_NEWS_DATA=Data_Berita
COLLECTION_NEWS_SUMMARY_DATA=Data_Berita_Ringkasan
```

4. Frekuensi Perbaruan Data

4.1 Laporan Keuangan

- Update per periode laporan keuangan (quarterly)
- Transformasi data setelah scraping selesai

4.2 Data Saham

- Update harian dari Yahoo Finance
- Historical data tersedia

4.3 Berita

- Update real-time/periodic scraping
- Ringkasan berita di-generate setelah scraping